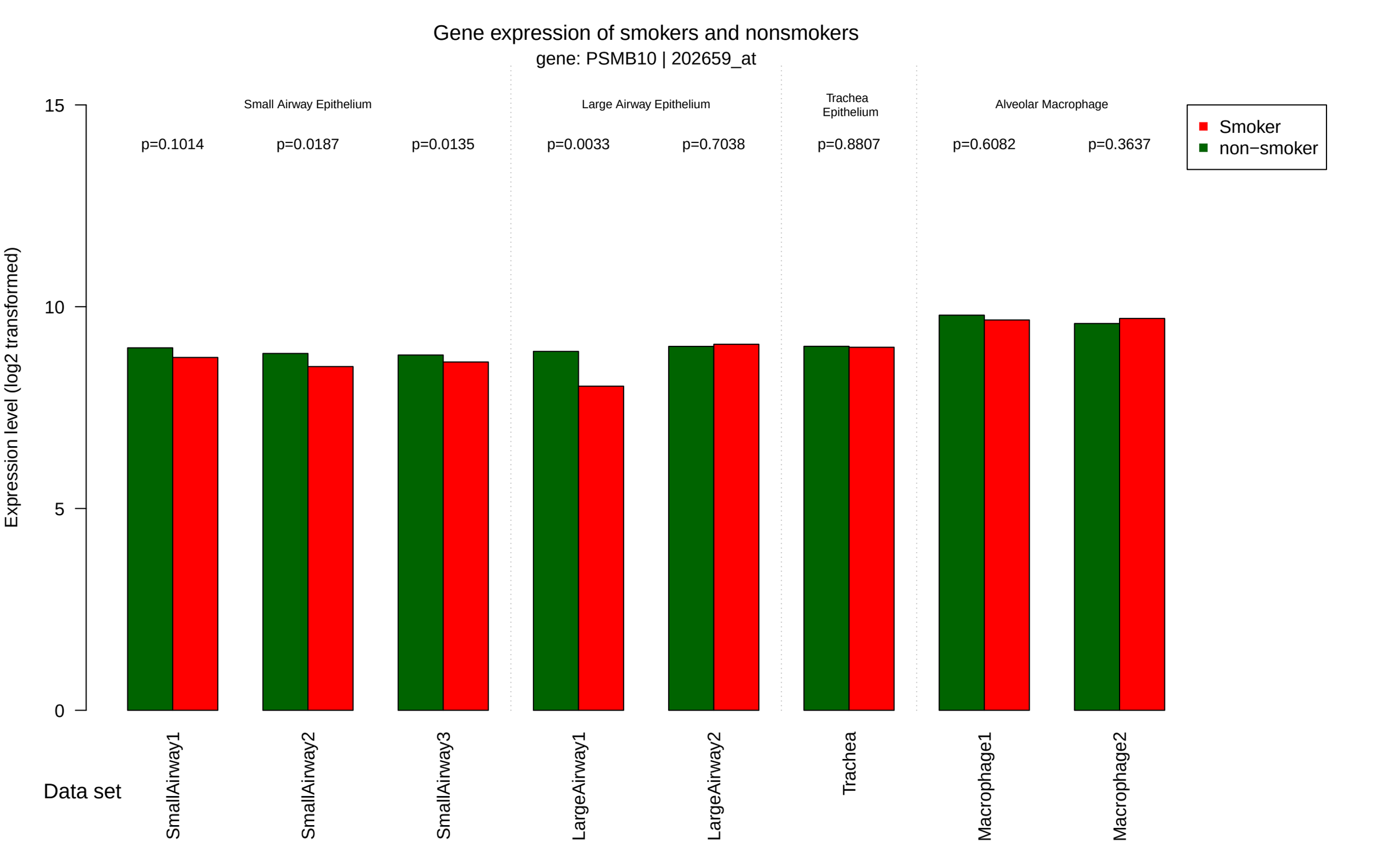
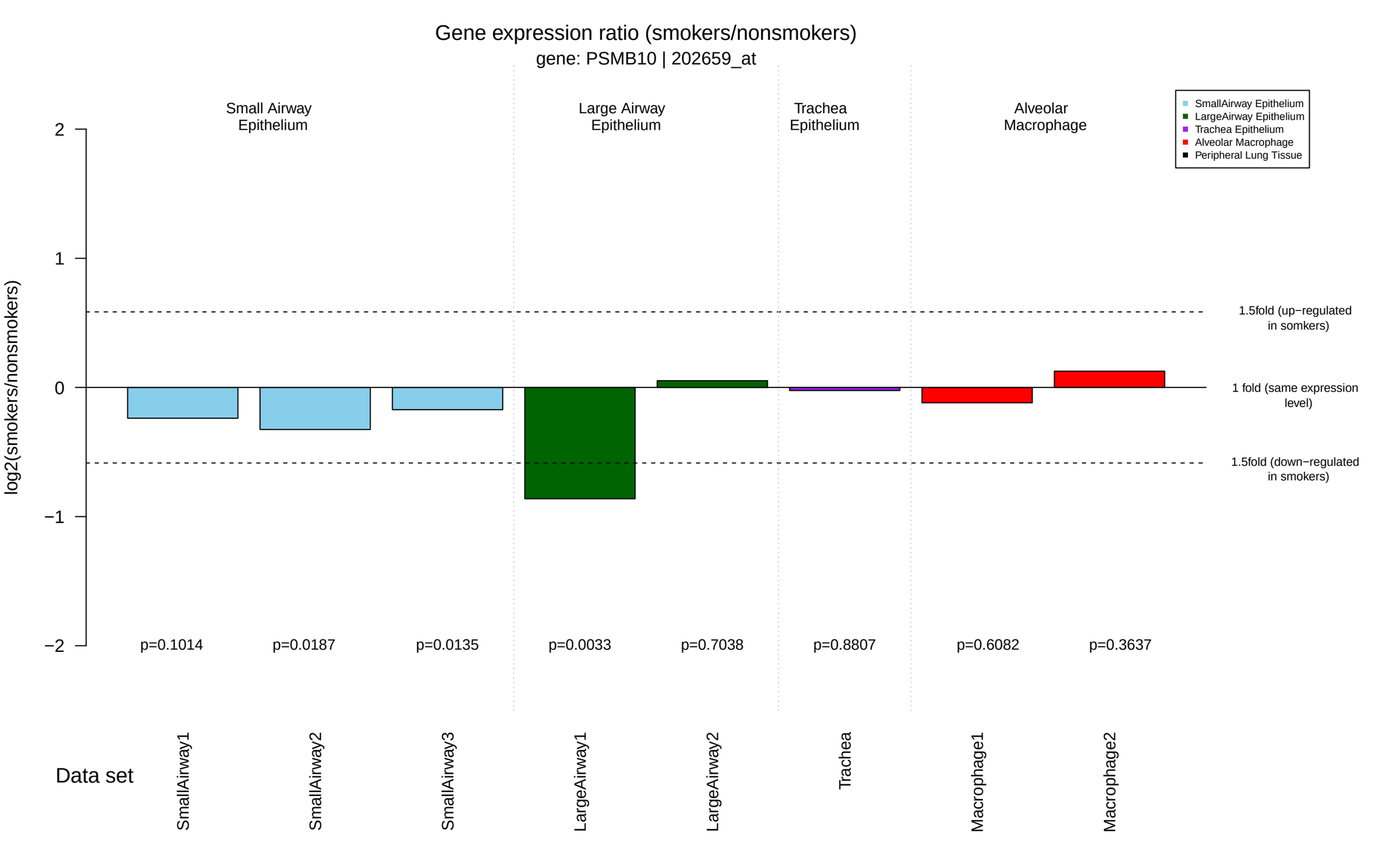
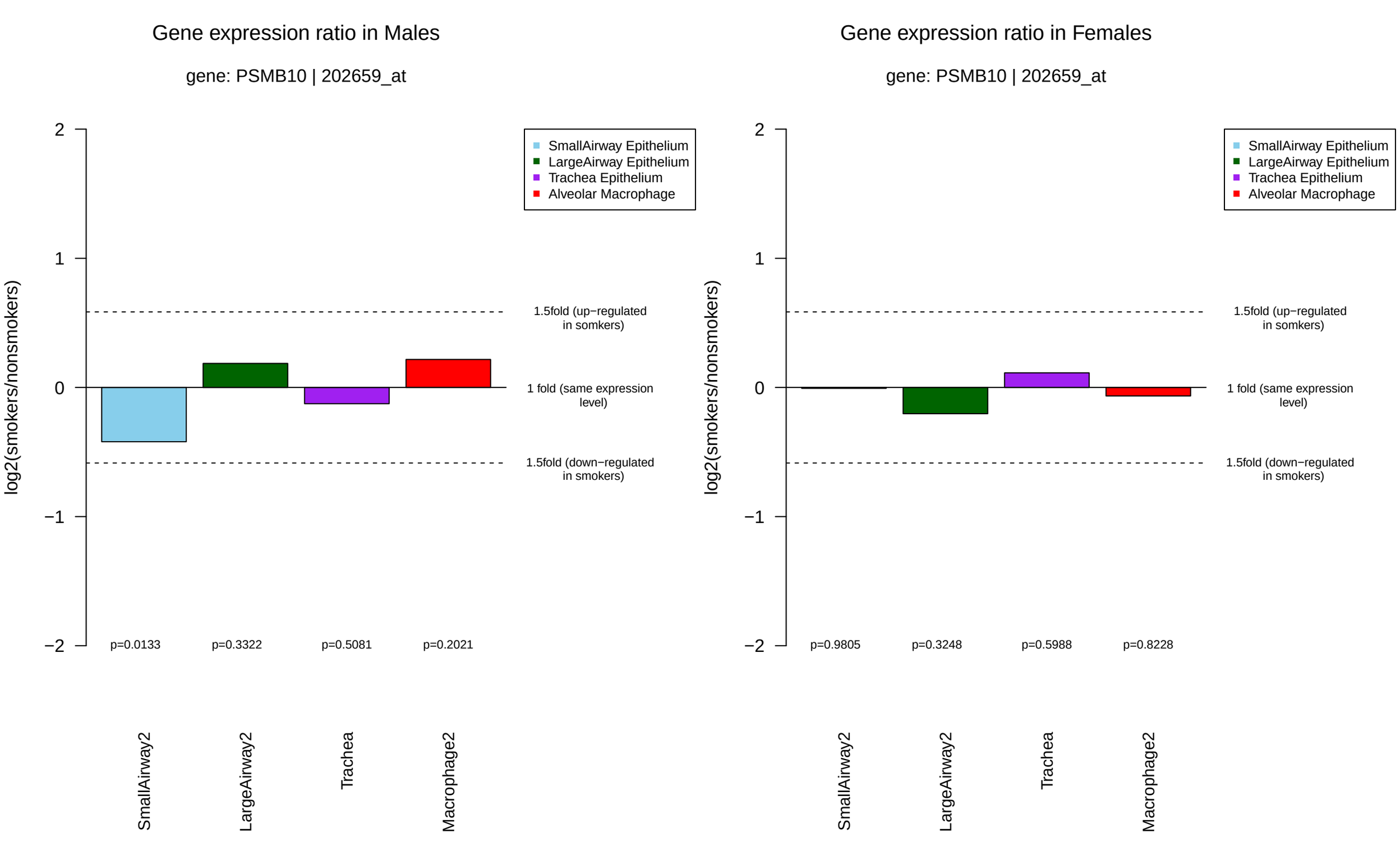
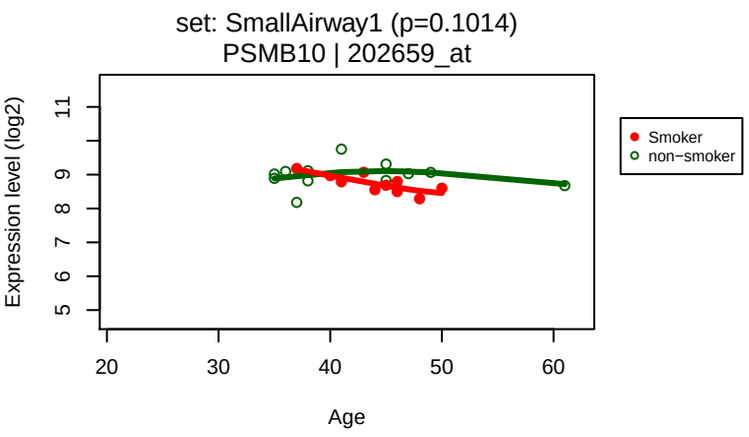
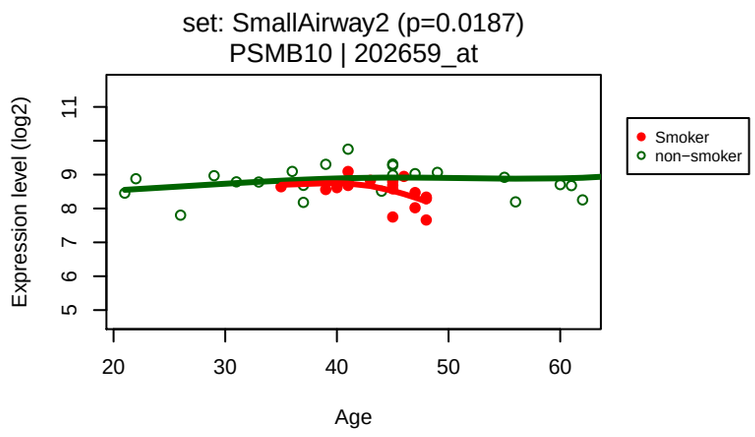
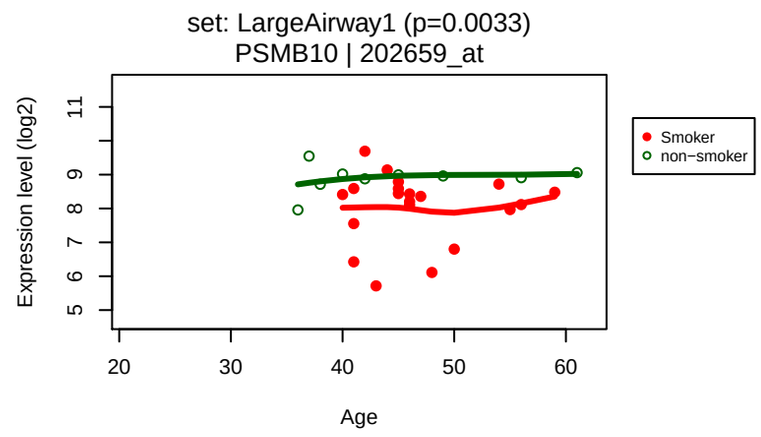

Gene expression of smokers and nonsmokers
gene: PSMB10 | 202659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1014
p=0.0187
p=0.0135
p=0.0033
p=0.7038
p=0.8807
p=0.6082
p=0.3637
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

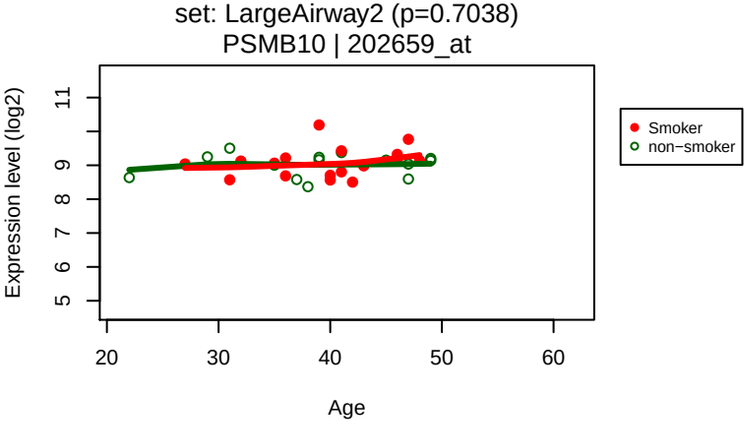

Gene expression ratio (smokers/nonsmokers)
gene: PSMB10 | 202659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1014
p=0.0187
p=0.0135
p=0.0033
p=0.7038
p=0.8807
p=0.6082
p=0.3637
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

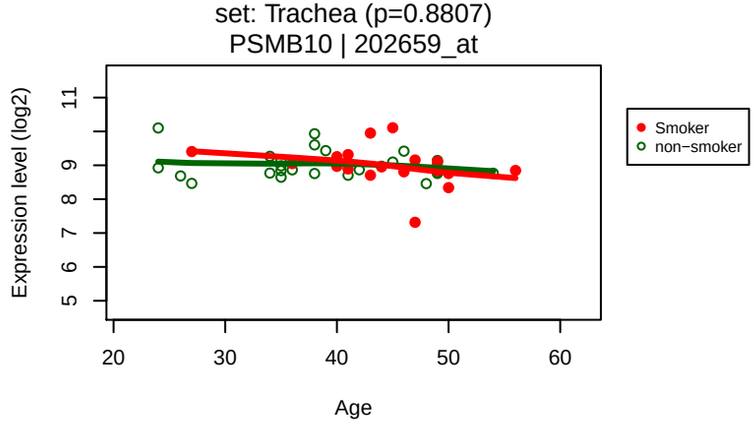

Gene expression ratio in Males
gene: PSMB10 | 202659_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0133
p=0.3322
p=0.5081
p=0.2021
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB10 | 202659_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9805
p=0.3248
p=0.5988
p=0.8228
SmallAirway2
LargeAirway2
Trachea
Macrophage2

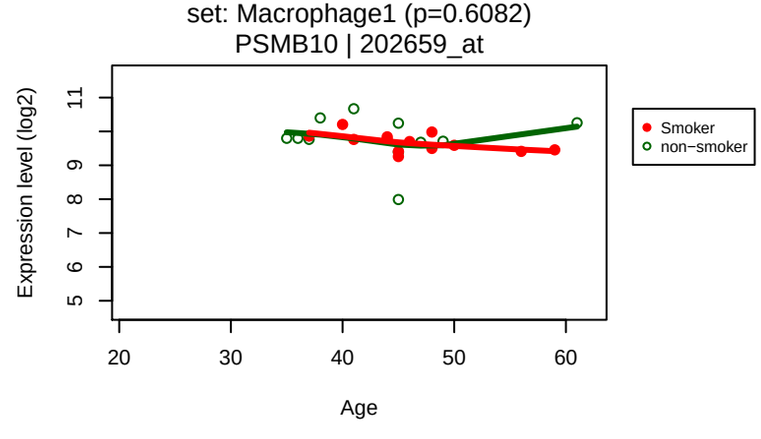

set: SmallAirway1 (p=0.1014)
PSMB10 | 202659_at
Expression level (log2)
Age
Smoker
non−smoker

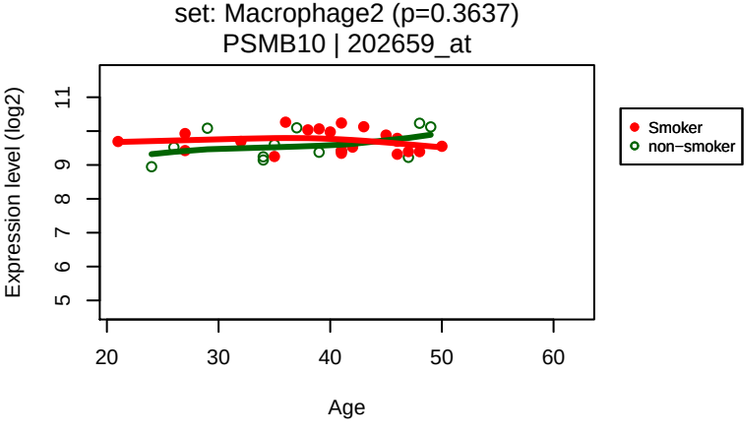

set: SmallAirway2 (p=0.0187)
PSMB10 | 202659_at
Expression level (log2)
Age
Smoker
non−smoker

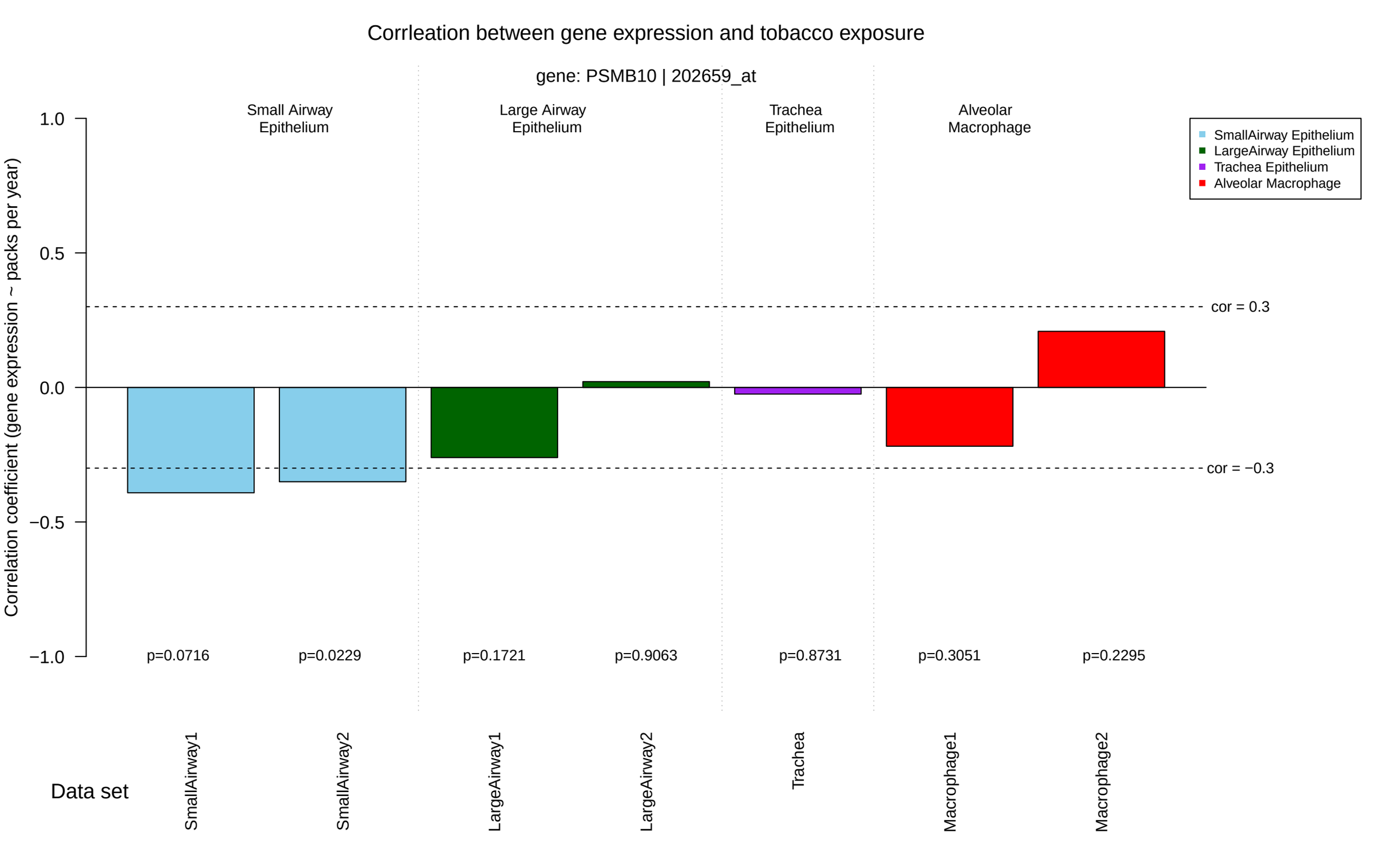

set: LargeAirway1 (p=0.0033)
PSMB10 | 202659_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7038)
PSMB10 | 202659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8807)
PSMB10 | 202659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6082)
PSMB10 | 202659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3637)
PSMB10 | 202659_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB10 | 202659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0716
p=0.0229
p=0.1721
p=0.9063
p=0.8731
p=0.3051
p=0.2295
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2